

IMS Hands-On Lab

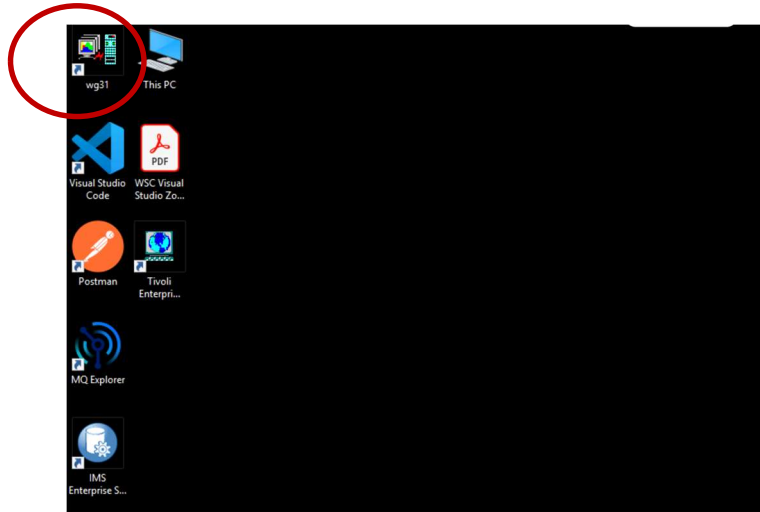
z/OS Connect and IMS OpenAPI 2

First things first

In this lab, the IMS Phonebook sample application (IVTNO) is used as the target IMS application program to expose through an API

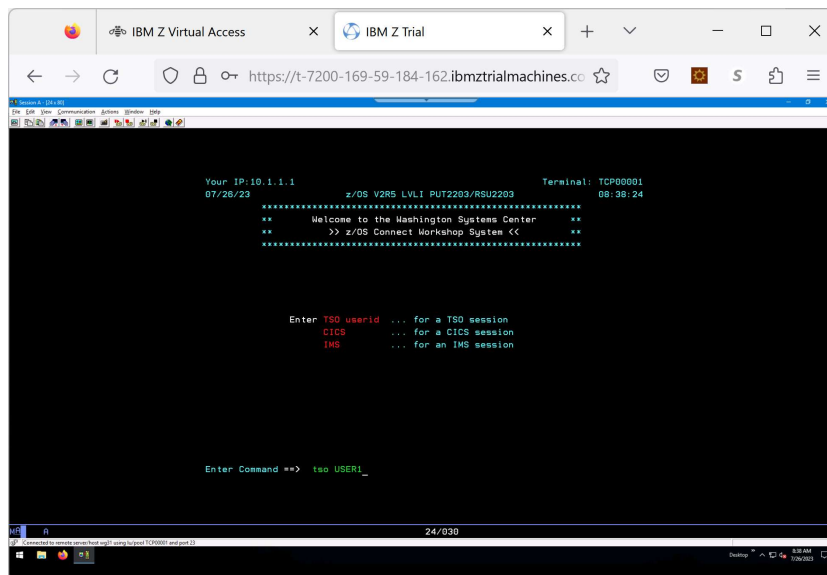
REVIEW the IMS Environment

Check out the IMS environment by double-clicking the **wg31 pcomm** icon

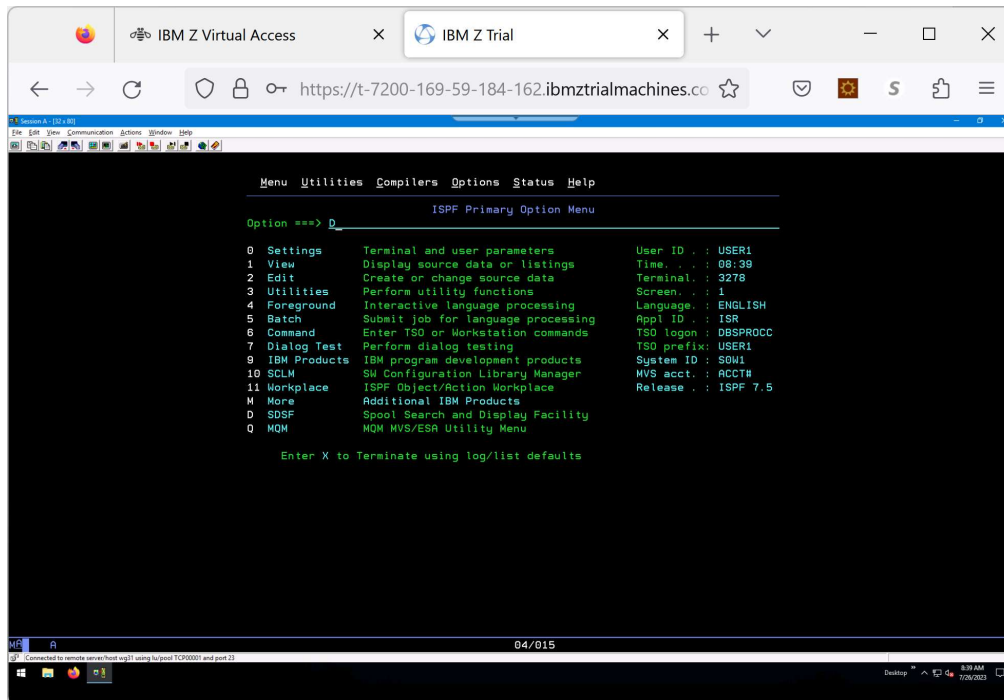


(note that shift- Enter is used as the enter function)

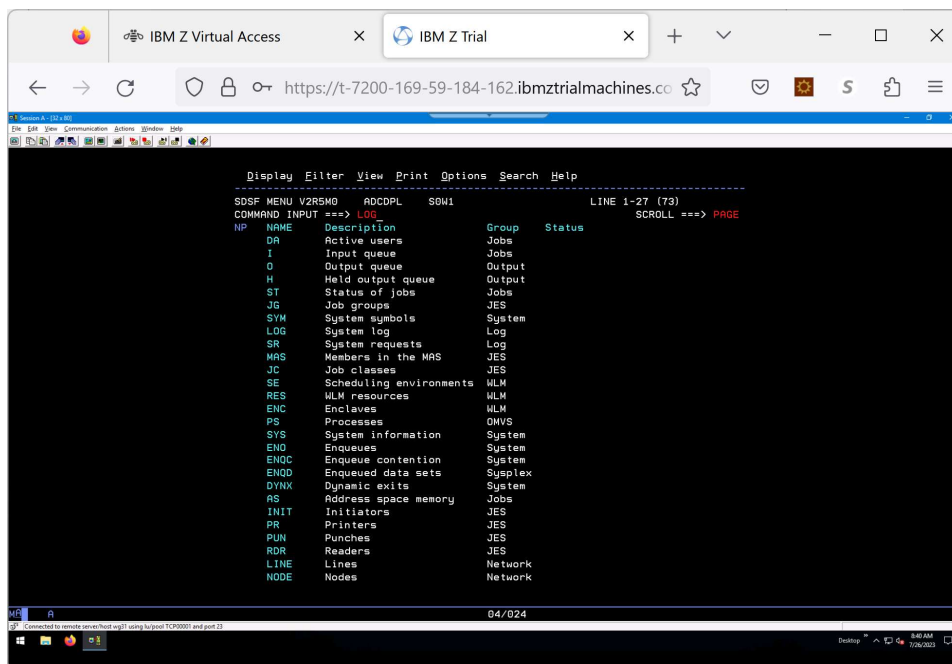
- Logon to TSO using USER1
 - Password is USER1



- ISPF option D allows access to SDSF



- Go to LOG



You will see the outstanding prompts for IMS (IVP1) and IMSConnect (IMS15HWS)

- Answer the outstanding prompt number (*nn*) for IMS (IVP1) to check on message region availability and the transaction ENQCT:

- **/nn/DIS TRAN IVTNO.**
-
- Make note of the value for ENQCT which will keep track of the number of accesses against the IVTNO transaction.

```

Display Filter View Print Options Search Help
-----
SDSF SYSLOG      0.101 SOW1 SOW1 07/26/2023 2W      4,784 COLUMNS 52- 131
COMMAND INPUT ==>
0290 R 05./DIS TRAN IVTNO.
0090 IEE000I REPLY TO 05 IS:/DIS TRAN IVTNO.
0090 DFS000I TRAN      CLS ENQCT  QCT  LCT  PLCT CP NP LP SEGSZ SEGNO
      PARLM  RC  IVP1
0090 DFS000I IVTNO      1      0      0 65535 65535 1 1 1 0 0
      NONE  0  IVP1
0090 DFS000I PSBNAME: DFSIVP1  IVP1
0090 DFS000I *23207/085309*  IVP1
0090 *07 DFS986I *IMS READY*  IVP1
S READY*  IVP1
IMS CONNECT READY* IMS15HWS
***** BOTTOM OF DATA *****

```

- You can also issue a display active command to verify that the dependent regions are activating
 - **/nn/DIS A.**

```

Display Filter View Print Options Search Help
-----
SDSF SYSLOG      0.101 SOW1 SOW1 07/26/2023 2W      4,785 COLUMNS 52- 131
COMMAND INPUT ==>
0090 DFS000I REGID JOBNAME TYPE  TRAN/STEP PROGRAM STATUS
      CLASS  IVP1
0090 DFS000I 1 IMSMSG1 TP
      1 IVP1
0090 DFS000I JWPGRN JMP  NONE  IVP1
0090 DFS000I JBPGRN JBP  NONE  IVP1
0090 DFS000I BRTCHRG BNP  NONE  IVP1
0090 DFS000I FRGRN  FP   NONE  IVP1
0090 DFS000I DBTRGN DBT  NONE  IVP1
0090 DFS000I IMSISGR1 DBRC      IVP1
0090 DFS000I IMSISDL1 DLS      IVP1
0090 DFS000I VTAM ACB OPEN -LOGONS ENABLED IVP1
0090 DFS000I IMSLUN=N/A N/A APPC STATUS=DISABLED TIMEOUT= 0
      MAYC= 5090 IVP1
0090 DFS000I DTMA GROUP=DTMAGRP STATUS=ACTIVE IVP1
0090 DFS000I APPC/OTMA SHARED QUEUE STATUS - LOCAL=INACTIVE
      GLOBAL=INACTIVE IVP1
0090 DFS000I APPC/OTMA SHARED QUEUES LOGGING=N IVP1
0090 DFS000I APPC/OTMA RRS MAX TCBS - 40 ATTACHED TCBS - 2 QUEUED
      RRSWKS= 0 IVP1
0090 DFS000I APPLID=IMS15 GRSHNAME= STATUS=DISABLED
      IVP1
0090 DFS000I TCP/IP GENINISID= STATUS=DISABLED IVP1
0090 DFS000I LINE ACTIVE-IN - 1 ACTIVE-OUT - 0 IVP1
0090 DFS000I MODE ACTIVE-IN - 0 ACTIVE-OUT - 0 IVP1
0090 DFS000I *23207/085407* IVP1
0090 *08 DFS986I *IMS READY*  IVP1
S READY*  IVP1

```

- Answer the outstanding prompt number (**zz**) for IMS Connect (IMS15HWS)

-
- The screenshot shows a web browser window with the address bar displaying 'https://t-7200-169-59-184-162.ibmztrialmachines.co'. The browser tabs include 'IBM Z Virtual Access' and 'IBM Z Trial'. The main content area shows a terminal window with the following text:
- ```

Display Filter View Print Options Search Help

SDFS SYSLOG 0.101 SOW1 SOW1 07/26/2023 2W 4,739 COLUMNS 02- 81
COMMAND INPUT ==> /Q3viewhws_ SCROLL ==> PAGE
S PARLM RC IVP1
N 4200000 SOW1 23207 08:48:39.44 STC01676 00000090 DFS0001 RC IVTNO 1
S NONE 0 IVP1
N 4200000 SOW1 23207 08:48:39.44 STC01676 00000090 DFS0001 PSBNAME:
N 4200000 SOW1 23207 08:48:39.44 STC01676 00000090 DFS0001 *23207/084B3
W 4200000 SOW1 23207 08:48:39.44 STC01676 00000090 *05 DFS996I *IMS READY*
4200000 SOW1 08.48.38 STC01676 *05 DFS996I *IMS READY* IVP1
8000000 SOW1 13.34.00 STC01685 *03 HMSC0000I *IMS CONNECT READY* IMS15HWS
***** BOTTOM OF DATA *****

```
- The terminal window also shows a status bar at the bottom with '04/031' and a connection message: 'Connected to remote server/host using lu6pool TCP0000I and port 23'.

IBM Z Virtual Access

https://t-7200-169-59-184-162.ibmztrialmachines.com/rdp/

Session A: (32 x 60)

File Edit View Communication Actions Window Help

Display Filter View Print Options Search Help

```

SDPF SYSLOG 0.101 SOW1 SOW1 07/26/2023 2M 4.759 COLUMNS 32-131
COMMAND INPUT ==> SCROLL ==> CSR
0000 HWSC00011 ODBM IMSPLEX MEMBER=IMS15HWS TARGET MEMBER=PLEX1
0000 HWSC00011 DATASTORE=IVP1 STATUS=ACTIVE
0000 HWSC00011 GROUP=OTMAGRP MEMBER=HWSMEM
0000 HWSC00011 TARGET MEMBER=OTMAMEM STATE=AVAIL
0000 HWSC00011 DEFAULT REROUTE NAME=HWSDEF IMS VERSION=15.1
0000 HWSC00011 RACF APPL NAME=IMSAPPL MULTIRTP= CASCAD=
0000 HWSC00011 HMA RCEE AGING VALUE=999999
0000 HWSC00011 OTMA ACK TIMEOUT VALUE=120
0000 HWSC00011 OTMA MAX INPUT MESSAGE=5000
0000 HWSC00011 SUPER MEMBER NAME= CMO ACK TOQ=
0000 HWSC00011 IMSPLEX=PLEX1 STATUS=NOT ACTIVE
0000 HWSC00011 MEMBER=IMS15HWS TARGET=PLEX1
0000 HWSC00011 NO ACTIVE ODBM
0000 HWSC00011 NO ACTIVE MSC
0000 HWSC00011 NO ACTIVE ISC
0000 HWSC00011 PORT=4000 STATUS=ACTIVE KEEPAV=0 NUMSOC=1
EDIT= TIMEOUT=0
0000 HWSC00011 NO ACTIVE CLIENTS
0000 HWSC00011 PORT=5555D STATUS=ACTIVE KEEPAV=0 NUMSOC=1
EDIT= TIMEOUT=6000
0000 HWSC00011 NO ACTIVE CLIENTS
0000 HWSC00011 NO ACTIVE RMTHCCON
0000 HWSC00011 NO ACTIVE RMTHICS
0000 DFS28B4I EXTERNAL TRACE DATASET DFSTRA01 FULL - SWITCHING TO DFSTRA02 .
IVP1
IMS CONNECT READY* IMS15HWS
S READY* IVP1
***** BOTTOM OF DATA *****

```

04/021

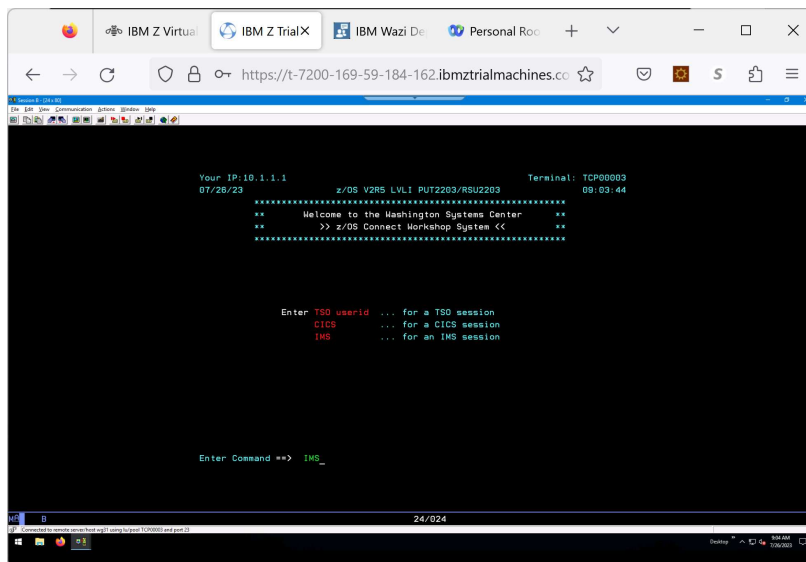
Connected to remote server host ug11 using lu6pool TCP0001 and port 23

8:51 AM 10/26/2023

- If PORT=4000 shows an inactive status, issue the command **/zzSTARTPT 4000**

It will be beneficial, prior to creating the API, to see what the transaction does and to test its use from a terminal (just to make sure everything is working).

Logon to IMS by opening up another pcomm session and double-clicking on the **wg31 Pcomm** icon on the the desktop



```

Your IP: 10.1.1.1 z/OS V2R5 LVL1 PUT2203/RSU2203 Terminal: TCP00003
07/26/23 * 09:03:44

** Welcome to the Washington Systems Center **
** >> z/OS Connect Workshop System << **

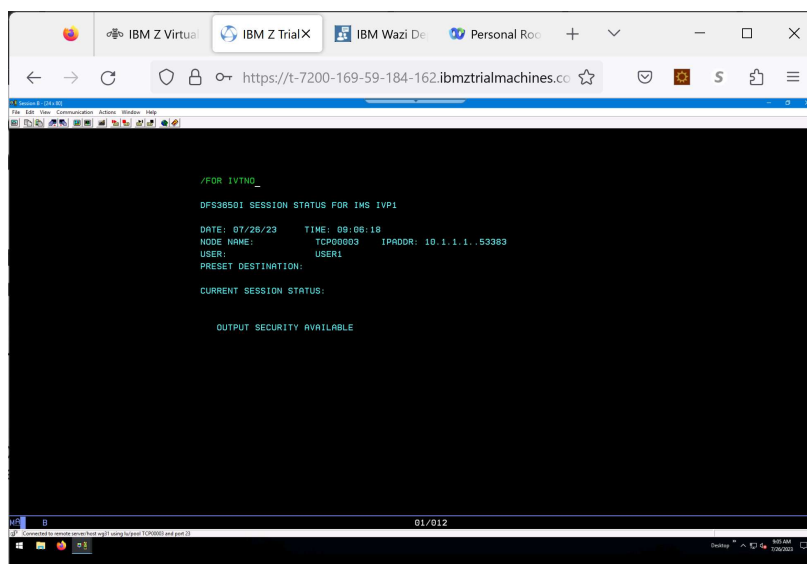
Enter TSO userid ... for a TSO session
CICS ... for a CICS session
IMS ... for an IMS session

Enter Command ==> IMS_

```

Userid and password are USER1/USER1

Once you are logged on, use format IVTNO (/FOR IVTNO)



```

/FOR IVTNO_
DFS3650I SESSION STATUS FOR IMS IVP1

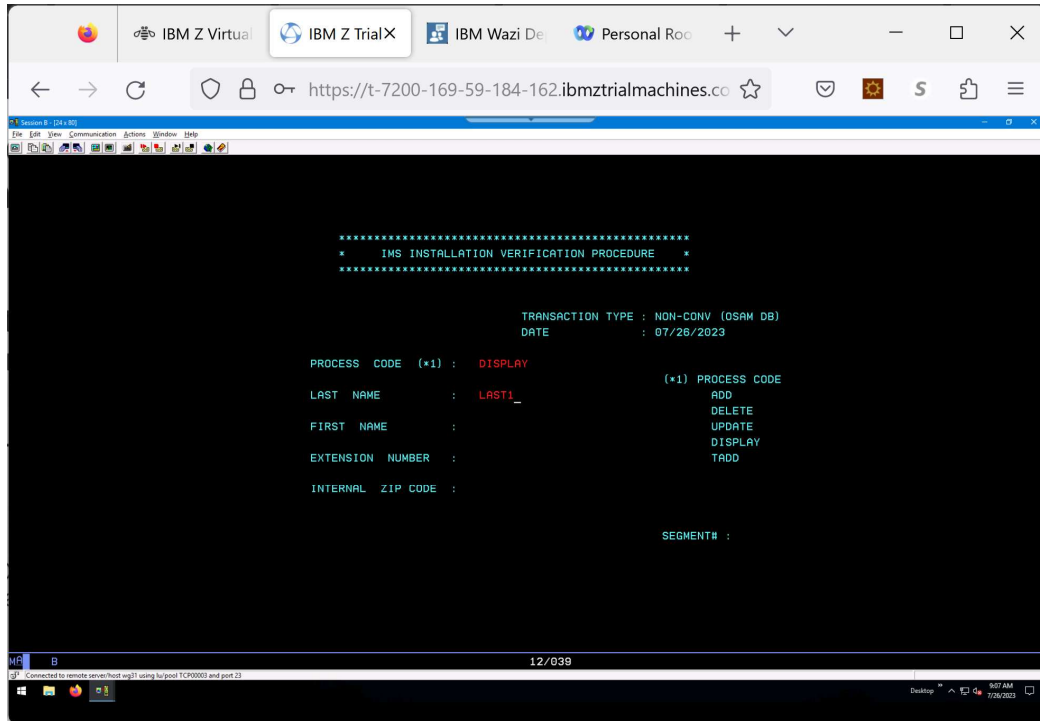
DATE: 07/26/23 TIME: 09:08:18
NODE NAME: TCP00003 IPADDR: 10.1.1.1..53383
USER: USER1
PRESET DESTINATION:

CURRENT SESSION STATUS:

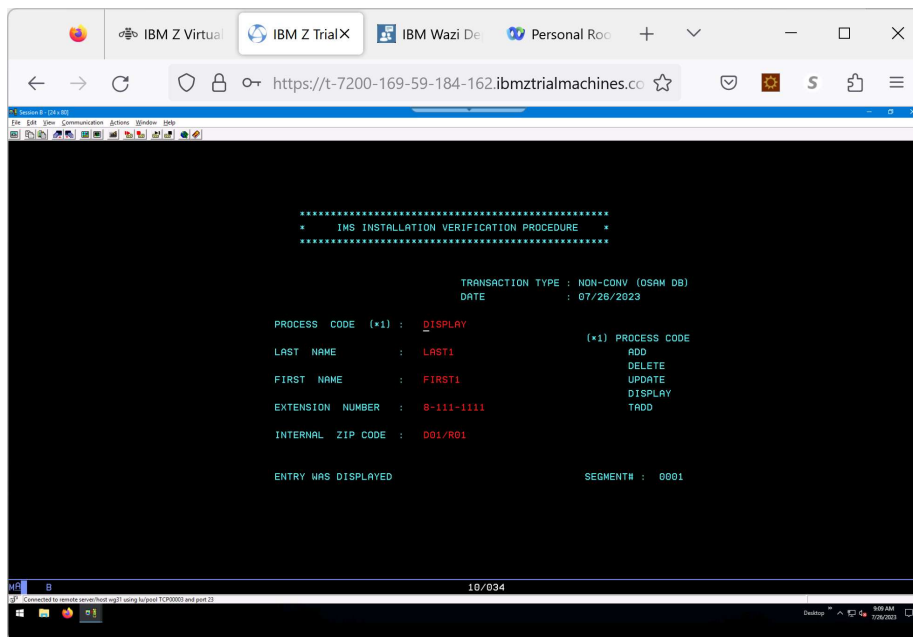
OUTPUT SECURITY AVAILABLE

```

Test the transaction using a process code of **DISPLAY** and a LAST NAME of **LAST1** (by the way, this is what we will be exposing using an API)



If everything on the IMS system is working properly, you will receive the results as follows



# HOW to encode your Userid/password for the API authentication

You won't need to do this for your lab since this has been done for you. We will use userid USER1 and password USER1.

This information is provided for your use in your own environment

Open up a browser and locate <https://www.base64encode.org/> and key in Userid:Password in the ENCODE window. Click ENCODE, and view the result in the bottom window. For our example,

The screenshot shows the website [www.base64encode.org](https://www.base64encode.org/). The main heading is "Encode to Base64 format" with the instruction "Simply enter your data then push the encode button." Below this is a large text input area containing "USER1:USER1", indicated by an orange arrow. Below the input area are several options: "UTF-8" for the destination character set, "LF (Unix)" for the destination newline separator, and checkboxes for "Encode each line separately", "Split lines into 76 character wide chunks", and "Perform URL-safe encoding". There is also a "Live mode" toggle which is currently "OFF". Below these options is a green button labeled "> ENCODE <" with the text "Encodes your data into the area below." to its right, indicated by an orange arrow. At the bottom, the encoded result "VVFNFUJE6VVFNFUJE=" is displayed in a light gray box, also indicated by an orange arrow.

When asked for authorization in the SWAGGER UI, code: BASIC VVFNFUJE6VVFNFUJE=